

REPORT ON WATER QUALITY INDICATORS FOR COMMUNITY-BASED MONITORING PLATFORM

ACADEMIC DEFENSE PROJECT – BAMENDA, CAMEROON

Prepared for: Academic Supervisor

Subject: Drinking Water Health Indicators and Proposed Monitoring Framework

1. Executive Summary

This report presents research findings on **drinking water health indicators** suitable for a community-based software monitoring platform in Bamenda, Cameroon. Following the supervisor's directive, only indicators that directly determine whether water is **healthy to drink** are included. Ecological indicators (Dissolved Oxygen, Electrical Conductivity, Temperature) and aesthetic-only indicators (colour, Odor) have been excluded. The selected indicators are scientifically valid (WHO 2024 guidelines), locally appropriate, and practically collectible by community members, water workers, and health officials.

2. Research Methodology

Online Sources:

- WHO Guidelines for Drinking-water Quality: Small Water Supplies (2024)
- Peer-reviewed studies on water quality in Bamenda, Cameroon
- Water Quality Index research for African contexts

Field Consultations:

- Water maintenance officer at Akom drinking water dam
- Community members in Bamenda regarding water security issues

3. Proposed Drinking Water Health Indicators

3.1 Physical Indicators (Community-Level Reporting)

Indicator	Reporting Method	Health Significance
Turbidity	Visual: "Clear," "Cloudy," "Very muddy"	High turbidity harbours pathogens and protects them from disinfection
Visible Focal Contamination	Direct observation: sewage, dead animals near source	Immediate indicator of cholera, typhoid, and diarrheal disease risk

3.2 Chemical Indicators (Water Worker-Level Testing)

Indicator	Health-Based Target	Testing Method	Health Significance
pH	6.5–8.5	Test strips or meter	Extreme pH causes tissue damage to mouth, oesophagus, and stomach
Nitrate (NO₃⁻)	< 50 mg/L	Test strips	Causes methemoglobinemia ("blue baby syndrome") – fatal in infants
Nitrite (NO₂⁻)	< 3 mg/L	Test strips	More toxic than nitrate; causes blood oxygen transport failure
Free Chlorine Residual	0.2–0.5 mg/L	DPD test kit	Ensures water remains pathogen-free from treatment to tap

3.3 Biological Indicators (Health Official Reporting)

Indicator	Detection Target	Health Implication
E. coli / Fecal Coliforms	Fecal contamination	Gold standard – any detectable E. coli makes water unsafe to drink
Total Coliforms	General bacterial contamination	Indicates treatment or pipe integrity failure
Reported Diarrhea Cases	Waterborne disease incidence	Direct health outcome; elevation indicates community-wide drinking water failure
Cholera Cases	<i>Vibrio cholerae</i>	Directly linked to unsafe drinking water
Typhoid Cases	<i>Salmonella typhi</i>	Endemic in areas with unsafe drinking water

3.4 Composite Indicator: Pass/Fail Framework

Safe to Drink: E. coli = 0 + Turbidity < 5 NTU + pH 6.5–8.5 + Nitrate < 50 mg/L

Unsafe to Drink: Any single parameter fails

4. Locations for Data Collection

Location	Source Type	Priority	Rationale
Akom Drinking Water Dam	Surface water	HIGH	Initial research site; serves significant population

Pipe-borne water points	Municipal supply	HIGH	Primary source for 65% of households
Boreholes (3 Sub-Divisions)	Groundwater	MEDIUM	Secondary source for 18.6% of population
Springs and streams	Natural sources	MEDIUM	Used by 8.1%; often unprotected
Household wells	Groundwater	LOW	Very high contamination risk from nearby latrines

Geographic Coverage: Bamenda I, Bamenda II, Bamenda III

5. Stakeholder Roles in Data Collection

Stakeholder	Health Indicators to Provide	Frequency
Water maintenance officers	pH, nitrate, nitrite, free chlorine, turbidity	Daily/Weekly
Community members	Turbidity (visual), visible faecal contamination	Real-time
Health officials	Diarrheal, cholera, typhoid cases; E. coli results	Weekly/Monthly
NGOs	Intervention records, chlorination status	As needed
Ministry of Water Resources	Regulatory compliance data	Monthly

Health Data-to-Action Pipeline:

Community Report (cloudy water + dead animal) → Dashboard Alert → Rapid E. coli test → Positive result → "UNSAFE TO DRINK" alert → Boil water advisory → Chlorination/repair → Follow-up confirmation

6. How This Platform Addresses Gaps

6.1 No Existing Platform in Cameroon

Research confirms no centralized drinking water health data platform exists in Cameroon. This would be a first-of-its-kind initiative.

6.2 Moving Beyond Hardware Limitations

Hardware-Only Approach	Your Software Platform
Measures only chemical parameters	Integrates E. coli + turbidity + nitrate + health outcomes
No coordination mechanism	Direct alerts when water becomes unsafe
Expensive to scale	Scales via existing mobile phones
Requires spare parts supply chain	Remotely updatable

7. Technical Recommendations

Data Validation Rules:

- **E. coli:** Any positive result → "UNSAFE TO DRINK"
- **Turbidity:** > 5 NTU → boiling advisory
- **pH:** < 6.0 or > 9.0 → health alert
- **Nitrate:** > 50 mg/L → warning for pregnant women and infants
- **Free chlorine:** < 0.2 mg/L → recontamination alert

Minimal Health Testing Kit for Water Workers:

- pH strips (range 4–10)
- Nitrate/nitrite combo test strips
- Free chlorine DPD test kit
- Turbidity tube
- E. coli rapid presence/absence test

8. Conclusion

Following the supervisor's directive to select only indicators concerning water quality healthy to drink, this report includes:

- **Community-collectible:** Turbidity (visual), visible fecal contamination
- **Worker-collectible:** pH, nitrate, nitrite, free chlorine, turbidity
- **Health official-collectible:** E. coli, total coliforms, diarrheal, cholera, typhoid cases

All selected indicators meet three criteria:

1. **Scientifically valid** – WHO 2024 guidelines
2. **Directly health-relevant** – Failure makes water unsafe to drink
3. **Actionable** – Triggers specific public health response

The research confirms that a software-based monitoring platform focusing on drinking water health indicators is both feasible and urgently needed in Bamenda, Cameroon.

9. Request for Approval

Based on the evidence presented, I respectfully request approval to proceed with the software-based monitoring platform approach.

Next steps upon approval:

1. Finalize health indicator pass/fail criteria
2. Design data collection forms for each stakeholder type
3. Develop prototype dashboard showing "SAFE" / "UNSAFE" status
4. Pilot testing at Akom drinking water dam

References

1. WHO. (2024). *Guidelines for drinking-water quality: small water supplies*. Geneva.
2. African Water Association. (2025). *State of Urban Household Drinking Water Security Situation in Bamenda North West Region, Cameroon*.
3. African Water Association. (2025). *Effectiveness and health risk assessment of drinking water from different sources treated by local household water treatment methods in Bamenda, Cameroon*.
4. ACECoR. (2025). *Adapted Water Quality Indices for Water Quality Monitoring in Africa*.